



NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 3

CANDIDATE
NAME

CLASS

BIOLOGY

9816/01

Paper 1

22 September 2021

Candidates answer on the Question Paper.

2 hour 30 minutes

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the space provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Section B	
Total	

This document consists of **18** printed pages.

[Turn over

Section A

Answer **all** the questions in this section.

- 1** You should read through the whole of the insert and **Question 1** carefully and then use the information provided to answer the questions.

The insert includes **Fig. 1.1 – 1.3** and **Table 1.1 - 1.2**.

- (a)** Suggest the role of cyanobacteria in the evolution of the circadian clock in some eukaryotic organisms.

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[3]

- (b)** Identify the photosynthetic pathway of plants that is likely to be most affected by the circadian clock and explain the importance of circadian rhythm in these plants.

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[3]

(c) With reference to Fig. 1.1 in the insert, explain the principles of homeostasis.

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[4]

(d) With reference to Fig. 1.2 in the insert,

(i) explain how the disruption of the circadian clock could result in depression.

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[3]

(ii) suggest reasons why nerve impulses only travel in one direction across the synapse.

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[2]

(e) With reference to Fig. 1.3 in the insert and your knowledge,

(i) describe the characteristics of a normal human prion protein.

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[3]

(ii) explain how prions are involved in the pathogenesis of human prion diseases.

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[2]

- (f) A t-test is conducted and the student proposed the null hypothesis:

there is no difference between the sleep hours of control group and CJD group.

With reference to the results shown in Table 1.1 and the formulae provided below, calculate the value of t and the number of degrees of freedom.

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \quad \nu = n_1 + n_2 - 2$$

key to symbols

s = standard deviation

$\bar{x}$ = mean

n = sample size (number of observations)

ν = degrees of freedom

Show your working.

value of t =

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number of degrees of freedom =

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[1]

- (g) Use Table 1.2 and your answers to (f) to decide whether the null hypothesis suggested by the student should be accepted or rejected.

Explain your answer.

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[2]

Section B

Answer **one** question in this section.

Either

- 3** Discuss the roles of B cells in immune response and immune tolerance. [25]

Or

- 4** Discuss the roles of enzymes in biological processes and biotechnological applications. [25]

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